

UW PICO 5.09

File: append_stderr.c

```

#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <fcntl.h>
#include <sys/wait.h>

#define MAX_ARGS 64
#define BUF_SIZE 1024

// Utility function to trim leading and trailing spaces
char *trim_spaces(char *str) {
    while (*str == ' ' || *str == '\t') str++;
    if (*str == 0) return str;
    char *end = str + strlen(str) - 1;
    while (end > str && (*end == ' ' || *end == '\t')) end--;
    end[1] = '\0';
    return str;
}

void execute_redirection(char *line) {
    char *stdout_file = NULL;
    char *stderr_file = NULL;
    int stdout_append = 0;
    int stderr_append = 0;

    // 1. Parse stderr Redirection (2>> or 2>)
    char *err_append_pos = strstr(line, "2>>");
    char *err_trunc_pos = strstr(line, "2>");

    if (err_append_pos) {
        stderr_append = 1;
        *err_append_pos = '\0';
        stderr_file = trim_spaces(err_append_pos + 3);
    } else if (err_trunc_pos) {
        stderr_append = 0;
        *err_trunc_pos = '\0';
        stderr_file = trim_spaces(err_trunc_pos + 2);
    }

    // 2. Parse stdout Append / Overwrite Redirection (>> or >)
    char *out_append_pos = strstr(line, ">>");
    char *out_trunc_pos = strchr(line, '>');

    if (out_append_pos) {
        stdout_append = 1;
        *out_append_pos = '\0';
        stdout_file = trim_spaces(out_append_pos + 2);
    }

```

^G Get Help
^X Exit

^O WriteOut
^J Justify

^R Read File
^W Where is

[Unknown Command:]

^Y Prev Pg
^V Next Pg

^K Cut Text
^U UnCut Tex

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int stderr_append = 0;

// 1. Parse stderr Redirection (2>> or 2>)
char *err_append_pos = strstr(line, "2>>");
char *err_trunc_pos = strstr(line, "2>");

if (err_append_pos) {
    stderr_append = 1;
    *err_append_pos = '\0';
    stderr_file = trim_spaces(err_append_pos + 3);
} else if (err_trunc_pos) {
    stderr_append = 0;
    *err_trunc_pos = '\0';
    stderr_file = trim_spaces(err_trunc_pos + 2);
}

// 2. Parse stdout Append / Overwrite Redirection (>> or >)
char *out_append_pos = strstr(line, ">>");
char *out_trunc_pos = strchr(line, '>');

if (out_append_pos) {
    stdout_append = 1;
    *out_append_pos = '\0';
    stdout_file = trim_spaces(out_append_pos + 2);
} else if (out_trunc_pos) {
    stdout_append = 0;
    *out_trunc_pos = '\0';
    stdout_file = trim_spaces(out_trunc_pos + 1);
}

// Clean base command token string
char *cmd_str = trim_spaces(line);
char *args[MAX_ARGS];
int arg_count = 0;
char *token = strtok(cmd_str, " \t");

while (token != NULL && arg_count < MAX_ARGS - 1) {
    args[arg_count++] = token;
    token = strtok(NULL, " \t");
}
args[arg_count] = NULL;

if (arg_count == 0) return;

pid_t pid = fork();
if (pid < 0) {
    perror("fork failed");

```

UW PICO 5.09

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} else if (out_trunc_pos) {
    stdout_append = 0;
    *out_trunc_pos = '\0';
    stdout_file = trim_spaces(out_trunc_pos + 1);
}

// Clean base command token string
char *cmd_str = trim_spaces(line);
char *args[MAX_ARGS];
int arg_count = 0;
char *token = strtok(cmd_str, " \t");

while (token != NULL && arg_count < MAX_ARGS - 1) {
    args[arg_count++] = token;
    token = strtok(NULL, " \t");
}
args[arg_count] = NULL;

if (arg_count == 0) return;

pid_t pid = fork();
if (pid < 0) {
    perror("fork failed");
    return;
} else if (pid == 0) {
    // Child Process: Redirect stdout if requested
    if (stdout_file && strlen(stdout_file) > 0) {
        int flags = O_WRONLY | O_CREAT | (stdout_append ? O_APPEND : O_TRUNC);
        int out_fd = open(stdout_file, flags, 0644);
        if (out_fd < 0) {
            perror("Failed to open stdout redirection file");
            exit(EXIT_FAILURE);
        }
        dup2(out_fd, STDOUT_FILENO);
        close(out_fd);
    }

    // Child Process: Redirect stderr if requested
    if (stderr_file && strlen(stderr_file) > 0) {
        int flags = O_WRONLY | O_CREAT | (stderr_append ? O_APPEND : O_TRUNC);
        int err_fd = open(stderr_file, flags, 0644);
        if (err_fd < 0) {
            perror("Failed to open stderr redirection file");
            exit(EXIT_FAILURE);
        }
        dup2(err_fd, STDERR_FILENO);
        close(err_fd);
    }
}

```

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^X Exit

^O WriteOut
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UW PICO 5.09

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} else if (pid == 0) {
    // Child Process: Redirect stdout if requested
    if (stdout_file && strlen(stdout_file) > 0) {
        int flags = O_WRONLY | O_CREAT | (stdout_append ? O_APPEND : O_TRUNC);
        int out_fd = open(stdout_file, flags, 0644);
        if (out_fd < 0) {
            perror("Failed to open stdout redirection file");
            exit(EXIT_FAILURE);
        }
        dup2(out_fd, STDOUT_FILENO);
        close(out_fd);
    }

    // Child Process: Redirect stderr if requested
    if (stderr_file && strlen(stderr_file) > 0) {
        int flags = O_WRONLY | O_CREAT | (stderr_append ? O_APPEND : O_TRUNC);
        int err_fd = open(stderr_file, flags, 0644);
        if (err_fd < 0) {
            perror("Failed to open stderr redirection file");
            exit(EXIT_FAILURE);
        }
        dup2(err_fd, STDERR_FILENO);
        close(err_fd);
    }

    execvp(args[0], args);
    perror("Execution failed");
    exit(EXIT_FAILURE);
} else {
    // Parent Process: Wait for process completion
    int status;
    waitpid(pid, &status, 0);
}

int main() {
    char input_buf[BUF_SIZE];
    while (1) {
        printf("\nAppend/Stderr Shell > ");
        if (!fgets(input_buf, sizeof(input_buf), stdin)) break;
        input_buf[strcspn(input_buf, "\n")] = 0; // Remove newline
        if (strcmp(input_buf, "exit") == 0) break;
        if (strlen(input_buf) == 0) continue;
        execute_redirection(input_buf);
    }
    return 0;
}

```

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```
medikondarakesh@Medikondas-MacBook-Air ~ % nano append_st.c
medikondarakesh@Medikondas-MacBook-Air ~ % nano append_stderr.c
medikondarakesh@Medikondas-MacBook-Air ~ % shell_complex.c
zsh: command not found: shell_complex.c
medikondarakesh@Medikondas-MacBook-Air ~ % nano shell_complex.c
medikondarakesh@Medikondas-MacBook-Air ~ % gcc -g -Wall shell_complex.c -o shell_complex
./shell_complex
```

```
Complex Shell > ls -la /invalid_folder /tmp &> combined.log
```

```
Complex Shell > cat combined.log
```

```
ls: /invalid_folder: No such file or directory
```

```
lrwxr-xr-x@ 1 root  wheel  11 24 Sep 12:40 /tmp -> private/tmp
```

```
Complex Shell > grep root < /etc/passwd | cut -d: -f1 > output_users.txt
```

```
Complex Shell > exit
```

```
medikondarakesh@Medikondas-MacBook-Air ~ % █
```